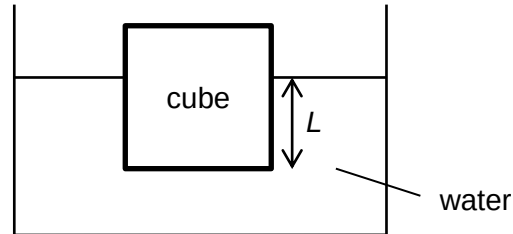


- 2** A cube of side 5.0 cm is floating on a tank of water as shown in the figure below. The density of the cube is 450 kg m^{-3} and the density of water is 1000 kg m^{-3} .



What is the submerged depth L of the cube?

A	2.3 cm	B	2.8 cm	C	4.0 cm	D	4.5 cm
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Answer: A

By the principle of floatation,
Weight of object = Weight of fluid displaced

$$V_c \rho_c g = V_w \rho_w g$$

$$V_c \rho_c = A L \rho_w,$$

$$L = \frac{V_c \rho_c}{A \rho_w} = \frac{0.050^3 \times 450}{0.050^2 \times 1000} = 0.023 \text{ m}$$

- 3** Water is ejected from the nozzle of a hose at a speed of 2.0 m s^{-1} . The density of water is